

## Challenge 2 — Reaching people quickly across a dispersed community

- **The problem.** The community is spread across a wide area, and when someone needs urgent help, the time it takes for the right response to reach them matters. Distance and dispersion work against speed.
- **Who it is for.** A resident who needs urgent help, and the people or services trying to reach them in time.
- **A good solution.** Shortens the gap between a need arising and the right response reaching the person, in a setting where distance slows things down.
- **For example.** A way to share a precise location and reach the nearest available help fast, a system that coordinates or routes responders more efficiently across a wide area, or a tool that helps a community respond in the first minutes before formal help arrives.

# 1 — Impact and value to the community

Emtad addresses fundamental challenges faced by rural communities such as Al Quaa where people, resources, and services are spread across large distances making it difficult to connect or to know they exist.

The beneficiaries are visitors/tourists, residents, local service providers, and camel farm owners. In many situations help may already exist nearby but community members/visitors don't know who's available or how to reach the right person asap which can lead to delays in responses to medical incidents, vehicle breakdowns, equipment failures, or other urgent situations.

Emtad reduces these delays by creating a trusted network and community where residents can request assistance or help those who request it and be matched with nearby people who have relevant skills, equipment, or resources. Whether they need veterinary support, vehicle recovery, water transportation, first aid assistance or locating a lost camel, the platform helps connect the person with helpful people/services or lead them to the solution and reduce the time taken to find someone or an emergency number.

Beyond that, the platform helps strengthen community communication through local announcements and updates.

Visibility and coordination is improved across the community, the platform helps build stronger ties and encourages collaboration between travellers, farms, residents, and makes a better use of resources that already exist within the area. The integrated reward system encourages participation.

## 2 - Relevance to the challenge

Emtad tackles one of the biggest challenges in rural communities: the first-response to emergencies of all kinds and degrees of intensity, reducing the time between an emergency occurring and the right help reaching the victim, going beyond a general emergency report application. This problem emerges due to the generally weak connectivity, whether it be low signals or the scarcity of networks nearby, amidst the endless dunes.

The solution is focused on creating a fast-track through each phase of the response; from **identification** of the issue → **re-directing** to the **right** services → **connecting** the victim with the responders closest to them.

Our platform creates a fast-track response by identifying the emergency, routing it to the appropriate service, and simultaneously connecting the victim with the nearest available responders. By minimizing verbal explanations, overcoming language barriers, reducing misunderstandings, and enabling offline communication, our solution helps communities respond faster while ensuring official emergency services are engaged when needed.

### 3 - Feasibility of implementation

Emtad is designed to be practical and deployable using existing technology and infrastructure. The platform primarily consists of a mobile application, cloud based backend services and offline services of which can be deployed using available technologies and integrating existing services like SMS messages, emergency calls, Fazaa for the reward system.

The solution doesn't require new physical infrastructure, specialized hardware, or major construction projects. It instead leverages resources that community members already possess, including smartphones, mobile internet connectivity, equipment, vehicles, local expertise. This reduces deployment costs and barriers.

The initial implementation can focus on a pilot deployment within Al Quaa, involving residents, camel farm owners, local service providers, and community volunteers. We'll add nearest hospitals, pharmacies, camel veterinarians, police stations, nearest towing service, fuel delivery, all including their contact numbers and locations. We'll extend that to where community volunteers can see when someone requests a service/emergency and then provide assistance for example when a car is stuck in the sand far away they can provide food and water, or assist with battery jump start, a tire change, someone getting lost, etc.

Key features such as assistance requests, location sharing, resource matching, community announcements, reputation score and rewards systems can be implemented using existing mapping and database technologies. We'll extend that to support Fazaa through the reward system and gain its benefits.

Potential challenges such as internet connectivity limitations have been implemented in the design.

Because the solution relies on existing software technologies, devices, and community participation rather than expensive infrastructure, it's realistic to deploy, maintain, and scale within Al Quaa and other rural communities with similar needs.

## National-Behavioural-Reward-Programme

The 'National Behavioural Reward Programme' aims to improve the quality of life in the UAE and incentivise positive behaviour through a point-based rewards system. Individuals who adopt positive behaviours can win points and use them to pay for government services.

The programme aims to empower the entire community and encourage it to adopt positive behaviours, support families, promote community interaction, encourage patriotism, ensure loyalty, spread good ethics and bolster the economy. It is the UAE's first programme in behavioural science which focuses on homeland, society and family, with focus on 90 behaviours.

It is one of the initiatives of the [Department of Behavioural Rewards of the Ministry of Possibilities](#) to support and carry out the strategic framework and the national model for behavioural rewards. It will be run in cooperation with [Fazaa](#), a social initiative which aims to develop social solidarity within the UAE - and several other partners in the private sector.

The National Behavioural Reward Programme has many initiatives, including:

- a Professional Diploma in Behavioural Economics- it aims to qualify and develop participants' skills in analysing and motivating positive behaviour in the community
- an app for the Behavioural Reward Programme (Fazaa) - The app intends to maximise the benefits of the programme by facilitating accessibility for the targeted segments of the community. The Fazaa app is available on [App Store](#) and [Google Play](#).

Read related news coverage on [WAM](#).

[@HSHskMohd](#) launches the "National Behavioural Reward Programme" at Qasr Al Watan in Abu Dhabi. [#UAE](#).

<https://t.co/elrk1w82tF> [pic.twitter.com/2pM7ktWsNX](#)

— Dubai Media Office (@DXBMediaOffice) April 17, 2021

Fig 1: Explains initiatives taken by the government to include a reward system that will run in cooperation with Fazaa to provide an incentive to boost positive behaviours.

## 4 - Readiness of the solution

This solution is designed to complement existing emergency services and community initiatives. While residents often rely on neighbours, volunteers, or social media groups for assistance in rural areas, there is currently no unified and reliable platform that connects victims, nearby community responders, and official emergency services in a coordinated way. EMTAD bridges this gap by providing a structured, location-aware response system that ensures help reaches the victim as quickly as possible.

Our prototype demonstrates the complete emergency response workflow:

Authentication → Emergency Reporting → Incident Classification → Smart Routing → Offline SMS Backup → Official Escalation (if required) → Community Support.

Link to our app workflow generation:

<https://app.visily.ai/projects/3683682a-b61a-4c2c-9610-f8e26c80617e/boards/2651577>

Although this is currently a high-fidelity interactive prototype, every major feature has been designed around realistic implementation using existing technologies. To address rural connectivity challenges, the app includes an offline SMS fallback that sends the victim's location and incident details when internet access is unavailable. Critical incidents can also be escalated directly to official emergency services, while smaller cases are directed to nearby verified community volunteers.

Alongside emergency response, the platform includes community features that help residents prepare for emergencies, connect with verified volunteers, and access safety resources. Although the current submission is a prototype, it demonstrates the complete user journey and is designed to be implemented using existing technologies such as GPS, SMS, mapping services, and UAE PASS authentication.

## 5 — Scalability after the hackathon

The solution delivered through Emtad can be expanded and adapted to other communities with different challenges all around the Emirate. While it is not location specific, it mainly applies to rural areas where help is hard to find. It is an easy to use application with a simple UI that is catered especially to the large population of elderly people and families with children residing in Al Quaa.

With the information provided by the user about their emergency situation, the app identifies an appropriate solution and alerts verified community volunteers, formal help and emergency contacts. The app is an independent platform that is not directly tied to an emirate specific app as well. This makes it highly flexible as it is adaptable to the unique needs of every individual user and can be expanded to a larger scale in the future.

## 6 — Falsifiability and evidence

From the evidence provided, it can be concluded that many rural areas are in need of a system that manages off road emergencies and have developed a community of volunteers willing to take care of these situations. While these communities provide valuable assistance, they are not well known, may have trouble with coordination, response time, and rely solely on informal community help. Our application combines community help, formal help, and includes features which negate the issues that volunteer groups face.

Evidence was found throughout multiple scattered platforms that aren't easily found when needed. SOS calls and volunteering communities + aid services were found scattered across Facebook groups and posts, Telegram channels, and Reddit posts.

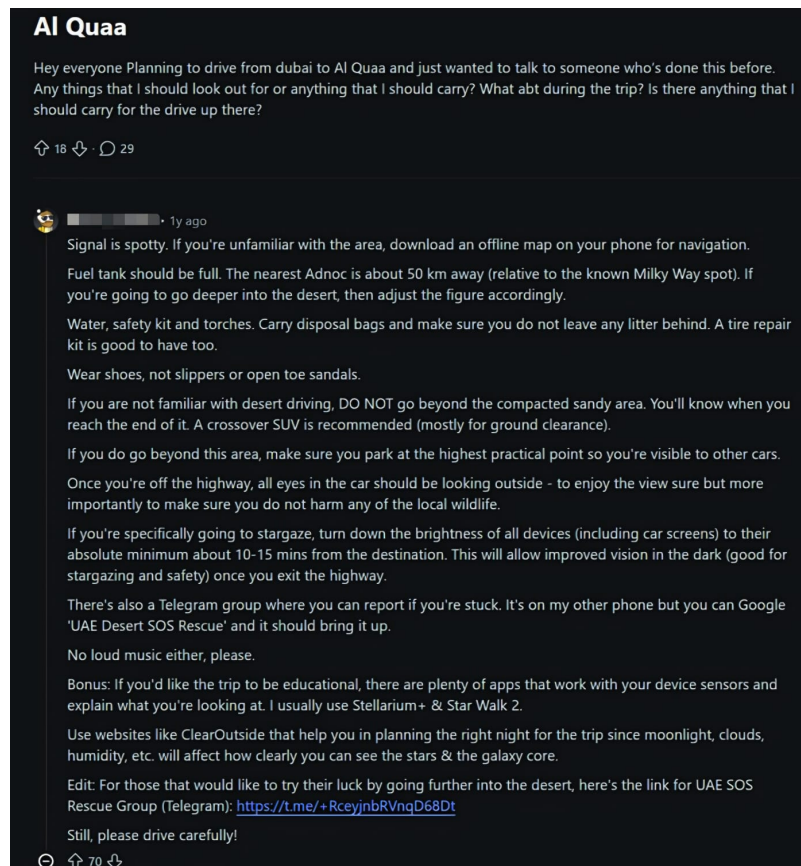


Fig 1: Talks about the dangers of travelling through Al Quaa and advice in case of emergency situations.

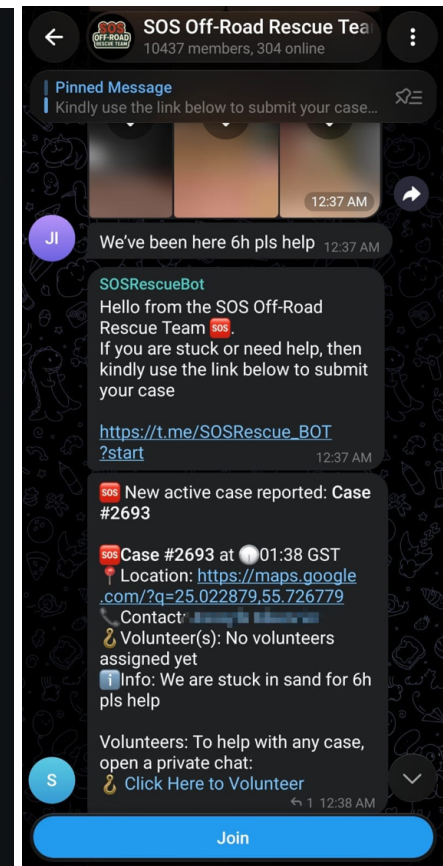


Fig 2: A telegram community volunteer group focusing on off road rescue.

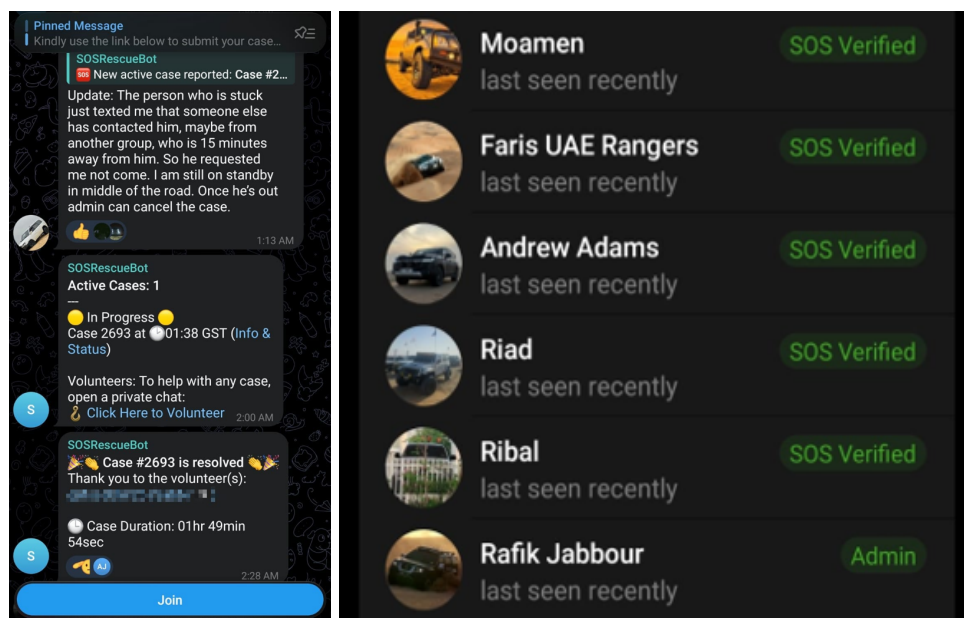


Fig 3: A better insight into a volunteer group's system and management.

Fig 4: Verified community personnel responsible for assisting people in distress



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Guys, just a word of caution on Al Quaa where we went yesterday. While the place is really good, please be very careful before venturing into sand. Anything just alongside the road also in terms of sand can be very tricky. Unless you are very sure about your car, driving skills and equipment, please stay on the hard part of the road. We had so many cars struck up there yesterday. Also while phone signals are there, internet connection generally doesn't work so difficult to send location. But the place itself is very nice and easily accessible by any car with excellent roads till the end

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Who's in Al Quaa now? Can you please help us? Our car got stuck in the sand.

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8 comments

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## 7 — References:

### **Behavioural Reward Systems**

<https://u.ae/en/about-the-uae/strategies-initiatives-and-awards/strategies-plans-and-visions/social-affairs/national-behavioural-reward-programme#:~:text=The%20'National%20Behavioural%20Reward%20Programme'%20aims%20to%20improve%20the%20quality.to%20pay%20for%20government%20services.>  
<https://www.wam.ae/en/details/1395302927656>

### **Evidence of community validation**

[https://www.reddit.com/r/dubai/comments/1kwmz17/al\\_quaa/](https://www.reddit.com/r/dubai/comments/1kwmz17/al_quaa/)  
<https://t.me/+RceyjnRVnqD68Dt>  
<https://www.facebook.com/abudhabirecoveryservice>

### **Geographical Insights**

<https://maps.app.goo.gl/cESXUPGPXH9C8BWT8>  
<https://www.city-facts.com/alqoua-al-ain-united-arab-emirates/population>